MILITARY SPECIFICATION

LOUDSPEAKER-MICROPHONE, (MEGAPHONE, ELECTRONIC, PORTABLE)

1. SCOPE

1.1 This specification covers the performance requirements for a transistorized megaphone, hand-held, single unit which shall be rugged, lightweight, portable, and shall be capable of amplifying the audio frequencies for transmitting intelligible speech over extended distances through high ambient noise levels. This specification covers essentially commercial type electronic megaphone equipment with minor modifications essential to adapt equipment for Military use.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-B-18 - Batteries, Dry.
MIL-I-983 - Interior Communication Equipment; Naval Shipboard; Basic Design Requirements for.
MIL-C-3965 - Capacitors, Fixed, Electrolytic (Tantalum), General Specification for.
MIL-E-17362 - Electronic Maintenance Parts Requirements Procedures for Stock Numbering and Assignment of Government Nomenclature.
MIL-P-17555 - Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-200 - Electron Tubes; and Semiconductor Devices, Diode.
MIL-STD-701 - Preferred and Guidance List of Transistors.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. - A specified number of samples, (see 6.1) of the loudspeaker-microphone, (electronic portable megaphone) equipment furnished under this specification shall be submitted to a laboratory designated by the bureau or agency concerned for performance of the tests specified herein.

FSC 5830

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3.2 General features.- The equipment shall be in accordance with the following paragraphs of Specification MIL-I-983 in addition to the requirements specified herein: (Whenever a requirement of Specification MIL-I-983 conflicts with a requirement of this specification, the requirements of this specification shall govern.)

General information	Electrical tapes
Definitions	Batteries
Stressed or vital parts	Switches (see 3.5.4.2)
Substitution of (equal or superior) materials	Fuses
Fungus-inert materials	Fuseholders and fuse clips
Unacceptable materials	Enclosure - degree of (see 3.5.3 and 3.5.3.1)
Acceptable materials	Minimum sheet metal thickness (watertight enclosures)
Flammable materials	Threaded devices
Arc-resistant materials	Rounded corners and edges
Toxic materials	Welding
Aluminum	Temperature and humidity
Iron or steel	Shock, vibration and inclination (vibration only - nonvital test required)
Brass	Shielding and radio frequency noise reduction
Bronze	Ground potential and grounding
Copper	Soldering
Copper-beryllium alloy	Electrical parts mounting
Copper-nickel alloy	Internal subassembly connection
Nickel-copper alloy (Monel)	Terminal boards, connectors and terminals
Copper-nickel-zinc (nickel silver and German silver)	Wiring
Zinc	Color coding
Other metals	Electrical insulation
Springs	Dielectric strength and insulation resistance clearances
Plastics	Drawings - general
Ceramics	Drawings - preliminary
Impregnating, filling, potting, embedding and encapsulating compounds	Drawings - working
Lubricants and lubrication	Schematic diagrams
Painting	Wiring diagram
Protection against corrosion	Drawing list
Bolts, machine screws, studs and nuts	Assembly drawings
Gaskets	Interchangeability and standardization
Locking devices	Designation and marking
Washers	Workmanship and general examination
Parts - electrical - general	
Variable resistors	
Transformers	
Relays	

3.3 Design features.- The design features of the equipment shall be as follows:

(a) Naturalness and intelligibility of speech under quiet conditions.
(b) Penetrating ability of speech under noisy conditions.
(c) Dependability of operation.
(d) Simplicity of design.
(e) Accessibility for servicing.
(f) Lightweight and compactness.
(g) Ability to withstand vibration.
(h) Ability to withstand high humidity.
(i) Ability to withstand extremes of temperature.

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3.4 Material and parts.- The quality of the material and parts used in the construction of this equipment shall be equal to that covered by specifications referenced in Section 2 of this specification. It is not desired that all such material be tested in accordance with referenced specifications in an individual case; however, the inspector may require such material tests whenever in his judgment it is necessary to ascertain that the quality of a material used meets the requirements of the specification. Where material is not in accordance with the requirement of specifications, it shall be clearly demonstrated that an improvement in operating characteristics or a saving in weight without sacrifice in reliability can be accomplished thereby, and that such substitutions do not preclude the subsequent use of Navy standard material in effecting repairs or replacements. Departures from referenced specifications shall be noted on the working drawings, and will thus be subjected to specific approval of the bureau or agency concerned.

3.5 Electric megaphone.- The megaphone shall consist of the following integral components:

(a) Microphone and microphone housing.
(b) Horn assembly.
(c) Amplifier.
(d) Batteries.
(e) Operating handle.

3.5.1 Weight and size.- The maximum weight and size shall be as follows:

Weight, 7 lbs.; length, 15 inches; diameter of reproducer bell, 12 inches.

3.5.2 Microphone housing.- The microphone housing is in the rear metal detachable portion of the megaphone assembly, and shall consist of the following parts:

(a) Gain control.
(b) Rubber mouthpiece.
(c) Microphone screen.
(d) Microphone element.

The microphone housing shall be watertight.

3.5.2.1 Gain control location.- The gain control shall be so located within the housing that the control knob is located below the microphone unit and easily accessible from either side.

3.5.2.2 Rubber mouthpiece.- A rubber mouthpiece shall be provided on the microphone end of the housing and shall be large enough to prevent physical contact between the operator's face and any metal portion of the microphone housing when talking into the megaphone.

3.5.2.2.1 Attachment of mouthpiece.- The mouthpiece shall be secured to the housing in a manner to allow easy removal. A holding groove on the housing to accommodate a projection on the mouthpiece, or vice-versa, is permissible. No screws or similar attaching devices shall be permissible.

3.5.2.3 Microphone screen.- The microphone face shall be protected by a metal screen.

3.5.2.4 Microphone.- The microphone shall be of the magnetic noise cancelling type.

3.5.2.4.1 Microphone cushion.- The microphone unit shall be cushioned in an acoustical absorbent material, such as sponge rubber, to damp out audio frequencies being transmitted through the metal horn and rear housing of the electric megaphone assembly to the microphone unit.

3.5.2.4.2 Microphone positioning.- The microphone shall be offset upward in the vertical plane from the coordinate axis of the horn driver assembly. The exact microphone location should be a compromise between decreasing the acoustical feedback and increasing the visibility of the operator while using the megaphone.

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3.5.3 Horn assembly.- The horn assembly shall consist of the following components:

(a) Horn unit.
(b) Magnet structure.
(c) Voice coil and diaphragm assembly.
(d) Driver head plate, and annular plug (if required).

3.5.3.1 Horn unit.- The horn unit may be either of a straight or re-entrant design.

3.5.3.2 Magnet structure.- The magnet structure shall be of the "W" type.

3.5.3.2.1 Air gap clearance.- Minimum air gap clearance on either side of the voice coil assembly shall be 0.007 inch.

3.5.3.3 Serviceability.- Removal of the magnet structure as a unit together with the voice coil and diaphragm assembly shall not expose the magnetic gap.

3.5.3.4 Diaphragm material.- The diaphragm shall be of a molded phenolic material.

3.5.3.5 Driver head plate and annular plug assembly.- The driver head plate and annular plug assembly is that part of the driver unit to which the magnetic structure is secured. The assembly shall be of molded phenolic material. The use of an annular plug is optional.

3.5.4 Operating and carrying handle.- The operating and carrying handle shall be a pistol grip type handle including a trigger and microswitch. The handle shall be secured to the megaphone unit as near as possible to the center of gravity. A suitable strap-carrying ring shall be provided as an integral part of the handle.

3.5.4.1 Trigger.- A spring-operated trigger type mechanism shall be used. The trigger shall be located on the operating handle so that the forefinger and middle finger of either hand may cover the trigger when grasping the handle during normal use.

3.5.4.2 Switch.- A microswitch, or equivalent, shall be used which, when depressed, shall place the megaphone in operation.

3.5.4.3 Drainage.- Facilities shall be provided to permit drainage of any moisture from the operating handle.

3.5.5 Amplifier enclosure.- The amplifier enclosure shall consist of a transistorized amplifier in a watertight enclosure.

3.5.5.1 Battery section.- The battery enclosure shall be a separate watertight unit and shall be readily removed from the battery connections.

3.5.5.2 Instruction plate.- An instruction plate explaining operating technique for the equipment shall be provided on the amplifier section.

3.5.5.3 Amplifier unit holding brackets.- Holding brackets shall be firmly attached to the amplifier section to serve as holding members for the audio amplifier unit. The amplifier shall be held in place with four-point suspension.

3.5.5.4 Amplifier parts.-

3.5.5.4.1 Capacitors.- Tantalum type capacitors shall be in accordance with Specification MIL-C-3965.

3.5.5.4.2 Transistors.- Transistors shall be selected in accordance with Standard MIL-STD-701.

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3.5.5.4.3 Rectifiers.- Rectifiers shall be chosen from approved types specified in Standard MIL-STD-200.

3.5.5.4.4 Amplifier unit battery contacts.- Beryllium copper, silver-plated spring contacts shall be used for battery connections.

3.5.5.4.5 Battery supply.- Dry cells, type BA30 in accordance with Specification MIL-B-18 shall be held separated and rigid by means of spring type clips or other approved method.

3.6 Electrical requirements.-

3.6.1 Input impedance.- The input impedance of the amplifier at 1000 cycles shall be determined by the contractor. The input impedance shall not vary more than plus or minus 20 percent over a frequency range of 400 to 5,500 cycles per second (c.p.s.). Production amplifier input impedance at 1000 cycles shall not vary more than plus or minus 20 percent from that value determined by the contractor.

3.6.2 Load impedance.- The load impedance that the horn unit presents to the amplifier shall be determined by the contractor. The average impedance of production horn units between the frequencies of 1250 and 1750 c.p.s. shall be within plus or minus 20 percent of that value determined by the contractor.

3.6.3 Rated input.- The rated input shall be the voltage output from the microphone unit obtained as specified in 4.3.2.

3.6.4 Full gain.- The gain of the amplifier required to produce maximum equipment output within the distortion limits allowed and with rated input from an acoustically remote source shall be considered full gain.

3.6.5 Usable gain.- The gain of the equipment with rated input from the megaphone microphone and with the megaphone gain control set just before feedback shall be considered usable gain.

3.6.6 Gain control operation.- The gain control specified in 3.5.2.1 shall provide a logarithmic attenuation in output with a linear rotation of the control.

3.6.7 Rated output pressure.- At the full gain of the amplifier, the horn unit shall be capable of delivering an output of 112 decibels (db) (db with reference to 0.0002 dynes per square centimeter (dynes/cm.^{2})), when measured as specified in 4.3.3.

3.6.8 Usable output pressure.- At the usable gain of the equipment, the horn unit shall be capable of an output of 112 db (db with reference to 0.0002 dynes/cm.^{2}), when measured as specified in 4.3.4.

3.6.9 Amplifier distortion.- The percent harmonic distortion in the amplifier output shall not exceed 10 percent at the usable output pressure and shall not exceed 15 percent at the rated output pressure when measured as specified in 4.3.5.

3.6.10 Output sound pressure distribution.- The maximum deviation from on-axis sound pressures shall be 3 db at 15 degrees and 6 db at 30 degrees, when measured as specified in 4.3.6.

3.6.11 Frequency response.- The equipment frequency response shall fall with the limits shown on figure 1, when measured as specified in 4.3.7.

3.6.12 Noise and microphonics.- The internally generated noise in the amplifier shall be 55 db below rated output, when measured as specified in 4.3.8.

3.6.13 Loudspeaker noises.- The loudspeaker design shall be such that natural resonances of the mechanical structure shall occur at frequencies outside the useful range of the loudspeaker. In any event, loudspeaker noises such as buzzes and rattles, should be low enough in intensity so as not to impair the articulation or quality of speech transmission.

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3.6.14 Equipment operating time.- The equipment shall be capable of transmitting 3000-ten-second messages when cycled 10 seconds "on" at rated output and 50 seconds "off".

3.7 Identification plates.- Identification plates shall be furnished on the megaphone unit.

3.8 Technical manuals.- One copy of the technical manual shall be supplied and packed with each equipment. The technical manual shall be the commercial instruction book and repair parts list.

3.9 Equipment maintenance parts.- Equipment maintenance parts shall be in accordance with Specification MIL-E-17362.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 The inspection procedures to be performed on the equipment shall be as follows: (The attention of the manufacturer is invited to the applicable paragraphs of Specification MIL-I-983 for information concerning tests such as where conducted, failures, and test conditions.)

4.2.1 Preproduction inspection at a designated laboratory.- Preproduction tests shall be conducted at a laboratory designated by the bureau or agency concerned (see 6.1). When preproduction inspection is conducted at the manufacturer's plant or at other than a Government laboratory, they shall be performed or witnessed by the Government inspector. Sample(s) as specified in the contract or order of the loudspeaker-microphone electronic portable megaphone equipment shall be subjected to the following examination and tests which shall be performed in general in the order listed.

Examination and tests	Requirement paragraph	Test paragraph
General examination	3.2	MIL-I-983
Operating	3.2	4.3.1
Temperature and humidity	3.2	MIL-I-983
Enclosure (watertight)	3.2	MIL-I-983
Vibration (nonvital vibration test required)	3.2	4.3.10
Rated input and output pressure	3.6.3 and 3.6.7	4.3.2 and 4.3.3
Usable output pressure	3.6.8	4.3.4
Amplifier distortion	3.6.9	4.3.5
Output sound pressure distribution	3.6.10	4.3.6
Frequency response	3.6.11	4.3.7
Noise and microphonics	3.6.12	4.3.8
Loudspeaker noises	3.6.13	4.3.9

4.2.1.1 Acceptance-preproduction inspection.- The results of each test shall be compared with the specification requirements. In the event of failure to conform to this specification, the contractor shall correct the cause of failure in the preproduction samples and on future production units.

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4.2.2 Acceptance inspection.- A sample number of equipments shall be selected in accordance with 4.2.2.1.2, 4.2.2.1.3 and 4.2.2.1.4, and each of the selected equipments shall be subjected to the following group A, B, and C examination and tests. Groups A and B examination and tests shall be at the place of manufacture. Group C tests shall be conducted at a laboratory designated by the bureau or agency concerned. The tests shall be performed in general in the order listed.

Examination and tests	Requirement paragraph	Test paragraph
Group A		
General examination	3.2	MIL-I-983
Operating	3.2	4.3.1
Group B		
Rated input pressure	3.6.3	4.3.2
Rated output pressure	3.6.7	4.3.3
Usable output pressure	3.6.8	4.3.4
Frequency response	3.6.11	4.3.7
Noise and microphonics	3.6.12	4.3.8
Loudspeaker noises	3.6.13	4.3.9
Group C		
Amplifier distortion	3.6.9	4.3.5
Output sound pressure distribution	3.6.10	4.3.6
Enclosure (watertight)	3.2	MIL-I-983
Vibration (nonvital vibration test required)	3.2	4.3.10

4.2.2.1 Sampling for acceptance inspection.-

4.2.2.1.1 Inspection lot.- All equipment presented for acceptance inspection at one time shall be considered a lot. The lot may include the entire contact quantity or it may be the production of any convenient time period at the choice of the inspector. Prior to presenting the lot for inspection, the contractor shall have subjected each equipment to the group A and B examination and tests, and shall include in the lot presented for inspection only those equipments that have passed these tests.

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4.2.2.1.2 Sampling for group A examination and tests.- A sample of equipment shall be selected from each inspection lot in accordance with table I and shall be subjected to the group A examination and tests specified in 4.2.2 with lot acceptance based on the sampling requirements of Standard MIL-STD-105.

Table I - Sampling for group A examination and tests
AQL (approximately) = 1.5 percent defective.

Number of equipments in inspection lot	Number of equipments in sample	Number of equipments nonconforming on any group A examination and test	
		Acceptance number	Rejection number
7 and under	All	-	-
8 to 15	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

4.2.2.1.3 Sampling for group B tests.- A sample of equipments shall be selected from each inspection lot by the inspector in accordance with table II, and shall be subjected to each of the group B tests specified in 4.2.2 with lot acceptance based on sampling requirements of standard MIL-STD-105.

Table II - Sampling for group B tests.

Number of equipments in inspection lot	Number of equipments in sample	Number of equipments nonconforming on any group B test	
		Acceptance number	Rejection number
3 and under	All	-	-
4 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 and over	25	2	3

4.2.2.1.4 Sampling for group C tests.- Group C tests will be required when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected by the Government inspector and shall be subjected to each of the group C tests specified in 4.2.2 with acceptance based on 4.2.2.1.6.

4.2.2.1.5 Acceptance - group A and B examination and tests.- The results of each examination and test shall be compared with the requirements of this specification. Failure to conform to the requirements of this specification for any test shall be counted as a defect, and the equipment shall be rejected. If the number of such nonconforming equipments in any sample exceeds the acceptance number for that

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sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for inspection, provided all equipments in the lot have been retested by the contractor for the test(s) causing rejection and all noncompliance corrected.

4.2.2.1.6 Acceptance - group C tests. - The results of each test shall be compared with requirements of this specification. In the event of failure to conform to the requirements of this specification for any group C test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

4.3 Test procedures. -

4.3.1 Operating test. - The operating test shall consist of the following:

(a) With the megaphone connected for normal operation, the trigger switch shall be operated.
(b) The gain control shall be advanced gradually until acoustical feedback begins. The control shall be backed off until the feedback stops.
(c) Speak into the microphone to determine if the set is operating.

4.3.2 Rated input. - Rated input shall be determined as follows: The megaphone unit shall be set up under free field conditions and an input applied to the microphone unit from an artificial voice, (calibrated to deliver a pressure of 100 dynes/cm.^{2} at 3/4 of an inch at a warbie frequency of 1250 to 1750 c.p.s.) placed at a distance of 1/16 of an inch and on the microphone axis. The output of the microphone unit measured across a resistance equal to the input impedance shall be considered rated input.

4.3.3 Rated output pressure. - With the megaphone unit set up under free field conditions and with rated input to the amplifier, amplifier gain control set to provide rated gain, supplied from an acoustically remote source and warbled through a frequency range of 1250 to 1750 c.p.s., the megaphone output shall be measured at a distance of 10 feet on the horn axis. The megaphone output shall be as specified in 3.6.7.

4.3.4 Usable output pressure. - The megaphone shall be set up under free field conditions and an input shall be provided to the megaphone microphone from an artificial voice calibrated to deliver a pressure of 100 dynes/cm.^{2} at 3/4 of an inch at a warble frequency of 1250 to 1750 c.p.s. placed at a distance of 1/16 of an inch and on the microphone axis. The megaphone gain control shall be at the usable gain position. The megaphone output shall be measured at a distance of 10 feet on the horn axis. The output shall be as specified in 3.6.8.

4.3.5 Amplifier distortion. - With rated input applied to the amplifier input and with the amplifier gain control set to provide rated gain, the distortion contained in the amplifier output across any resistive load between plus or minus 20 percent of rated load shall be as specified in 3.6.9 at any frequency between 400 and 5500 c.p.s. with rated input applied to the amplifier input, and with the amplifier gain control set to provide an output from the megaphone equal to usable output pressure, the distortion contained in the amplifier output across any resistive load between plus or minus 20 percent of rated load shall be as specified in 3.6.9 at any frequency between 400 and 5500 c.p.s.

4.3.6 Output sound pressure distribution. - With rated output pressure being obtained from the megaphone unit, pressure measurements shall be made at both 10 and 20 degrees off the megaphone sound axis. Four measurements shall be made at each angle, two on the horizontal plane on either side of the sound axis and two in the vertical plane above and below the sound axis. The maximum deviation from the on-axis pressure values shall be as specified in 3.6.10. The off-axis pressure shall be considered to be the arithmetical average of the four pressures measured at each of the two angles.

4.3.7 Frequency response. - The equipment shall be set up under free field conditions, and an input shall be provided to the megaphone microphone from an artificial voice calibrated to deliver a pressure of 141 dynes/cm.^{2} at 3/4 of an inch placed at a distance 1/16 of an inch in front of the microphone and on the microphone axis. With the input warbled through a frequency range of 1250 to 1750 c.p.s. and the

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megaphone output measured at a distance of 10 feet on the horn axis, the megaphone gain control shall be set so that the output measured is 106 db (db with reference to 0.0002 dynes/cm.^{2}). With the volume control set in this position, the input shall be varied over a frequency range of 400 to 5500 c.p.s., and the megaphone output measured to determine conformance with 3.6.11. The volume control shall then be adjusted to give an output of 99 db (db with reference to 0.002 dynes/cm.^{2}) with conditions as specified above, and with the input varied over the frequency range of 400 to 5500 c.p.s., the megaphone output shall also conform to 3.6.11.

4.3.8 Noise and microphonics.- The amplifier shall be energized with the amplifier gain control set to provide rated gain, with the amplifier input terminated with a resistor equal in value to the input impedance of the amplifier, and with the amplifier output terminated with a resistor equal in value to the megaphone load impedance, and the voltage developed across the output resistor shall be as specified in 3.6.12, when measured with a thermal type voltmeter.

4.3.9 Loudspeaker noises.- The loudspeaker shall be operated at usable output to determine conformance with 3.6.13.

4.3.10 Vibration.- The nonvital tests specified in Specification MIL-I-983 shall be conducted with the electric megaphone strapped to the respective testing machine in a manner to allow a minimum of relative motion, between the equipment and the machine.

4.4 Inspection procedures.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- The equipment, accessories, technical publications, and repair parts shall be preserved and packaged by level A or C; packed by level A, B or C and marked by levels A and B or C, as specified (see 6.1), in accordance with Specification MIL-P-17555. Method II shall apply for level A preservation and packaging of the equipment.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of preproduction samples (see 3.1).
(c) Laboratory where preproduction testing will be conducted (see 4.2.1).
(d) Quantity of items required.
(e) Quantity of technical manuals required and distribution.
(f) Whether unit stock number is to be entered on the identification plate (see Specification MIL-I-983).
(g) If pilot pack is required (see Specification MIL-I-983).
(h) Level of packaging and level of packing required (see 5.1).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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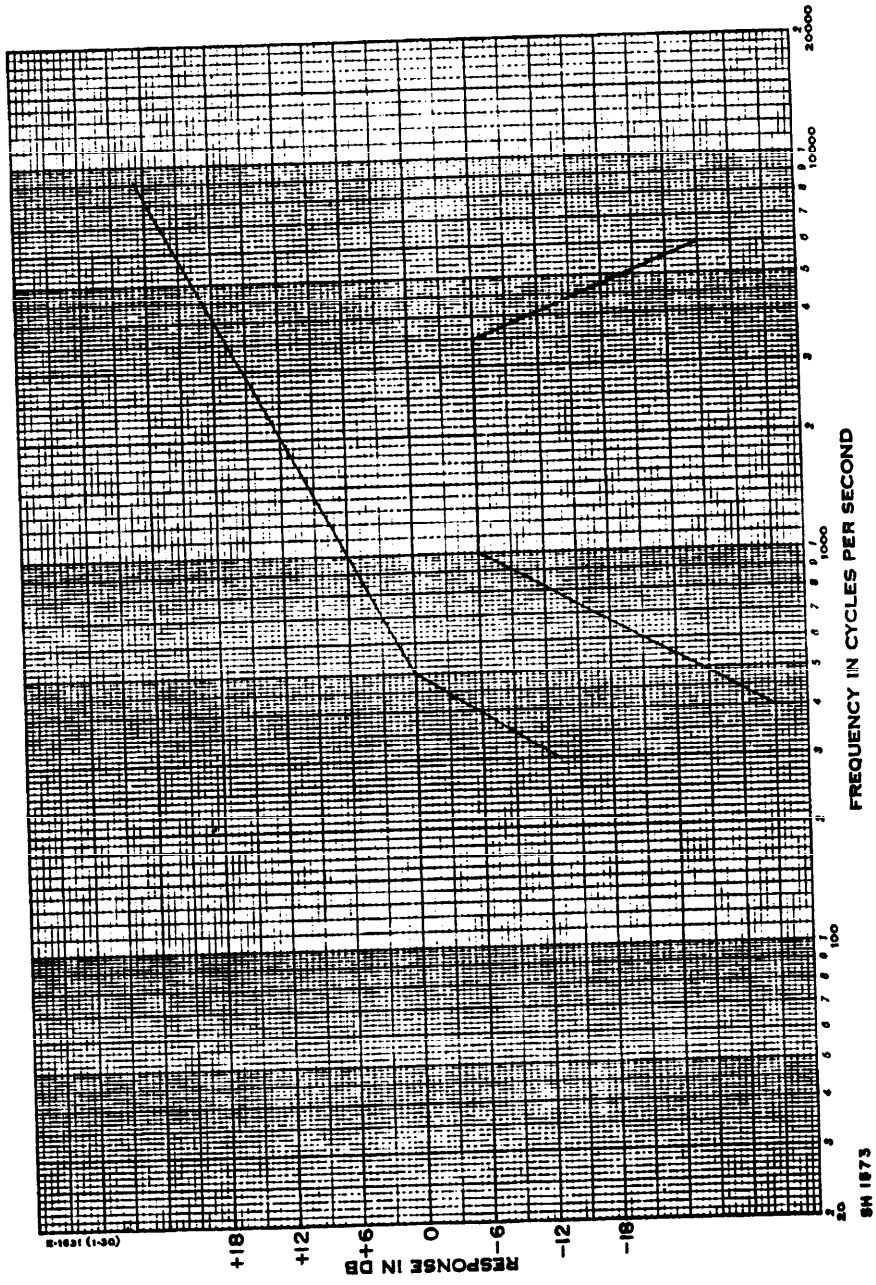


Figure 1 - System frequency response limits for electronic portable megaphone.

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